

Forecasting With Deep Temporal Hierarchies

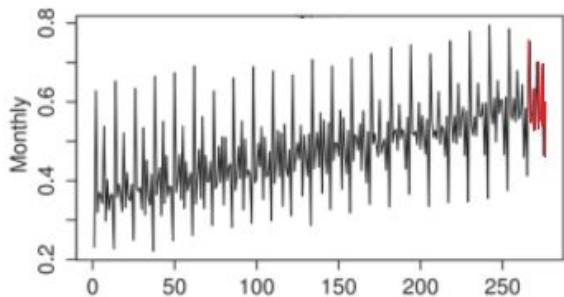
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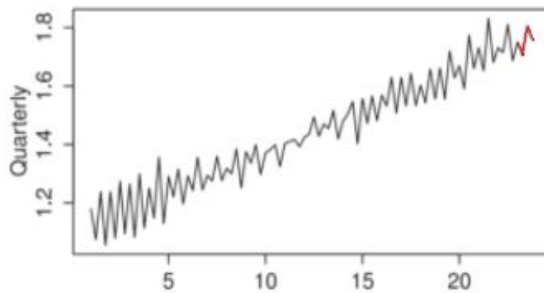
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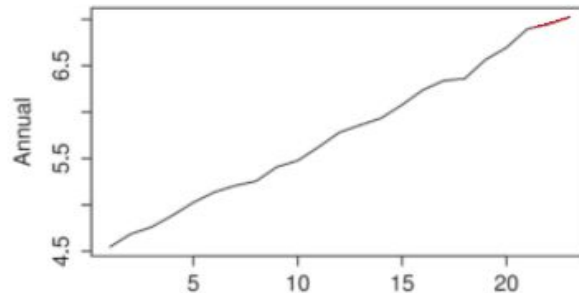
Introduction to Temporal Hierarchies



Seasonality Dominates



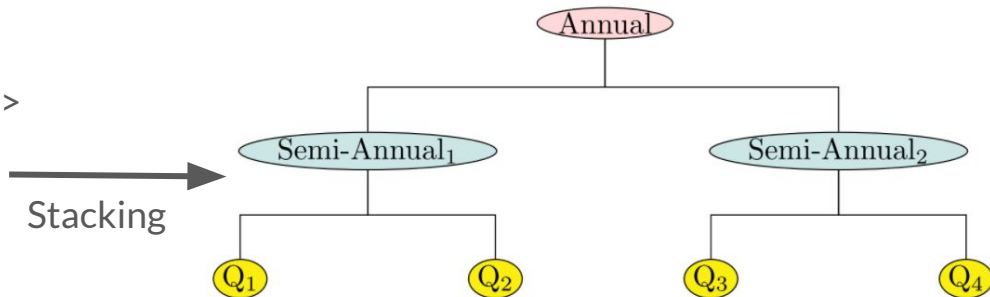
Both Seasonality + Trend



Clear Trend

Same Series ~ Different Viewpoints

- Trough non overlapping Temporal Aggregations =>
- Series of different frequencies =>
- Series with different information

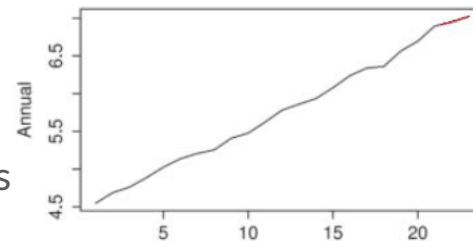
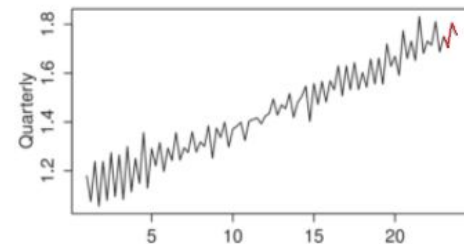
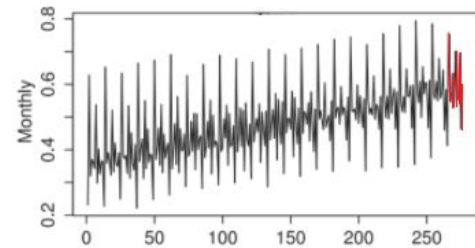


Why Temporal Hierarchies

- **Accuracy Improvements**
 - a. Combine Diverse Information
 - b. Parameter Estimation & Model Misspecifications
 - c. Aggregated Variance reduction:

$$\sigma = \sigma_1 + \sigma_2 + 2cov(\sigma_1, \sigma_2)$$

- **Aligned Decisions Between Hierarchical Levels**
 - a. Higher level strategic decisions $\simeq$ lower operational level decisions
 - b. Enhanced Decision Making



How to Forecast with Temporal Hierarchies

Temporal Hierarchical Forecasting - THieF (*Athanasopoulos et al, 2017*):

1. Get a **base forecast** for every aggregation level
2. **Reconcile** (Combine!!) them into the reconciled bottom level
3. **Sum - up** and reconstruct the entire hierarchy

Magic Formula

$$\tilde{\mathbf{y}}_h = \mathbf{S}\mathbf{G}\hat{\mathbf{y}}_h$$

How to reconcile?

- Analytical (econometric) methods : **Estimate G**
 - Linear methods
 - Assumptions to soften complex calculations
 - Partial - consideration of relationships between aggregation levels
- Machine Learning
 - Soft constraints **Coherence not ensured** (Mancuso et al. 2020; Mishchenko et al. 2019; Gleason 2020)
 - Local-Based Non-Linear methods : XGboost and Random Forest (*Spiliotis et al, 2020*)

Aim & Motivation

Introduce Deep Learning to the field of Temporal Hierarchies

1. Deep Learning for **Reconciliation** (Accuracy & Coherence)
2. Deep Learning for **Base Forecasts**
3. **Jointly** Forecast with Temporal Hierarchies (End - to - End)

Motivation: Combine the benefits of THieF and Deep Learning

- **Overcome limitations** of statistical methods through Deep Learning
 - Linearity
 - Assumptions
 - Partial Covariance Estimation
- Design Deep Learning architectures **based on THieF principles**

Methodology: NNs for Reconciliation part 1

Inspired by Encoders-Decoders & following THieF's principles

Model (**Reconciler**) Intuition:

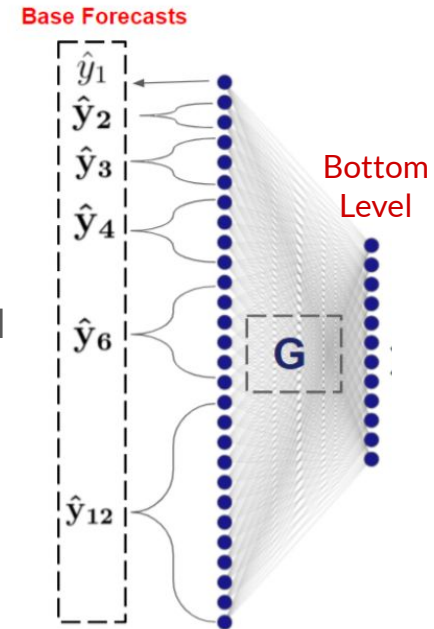
- Produce Base Forecasts (any model or non-model approach)
- Feed them into a Neural Network to combine them into the bottom level
- Construct the entire hierarchy



Key question:

What is the Optimization methodology!?

How to actually train the model?

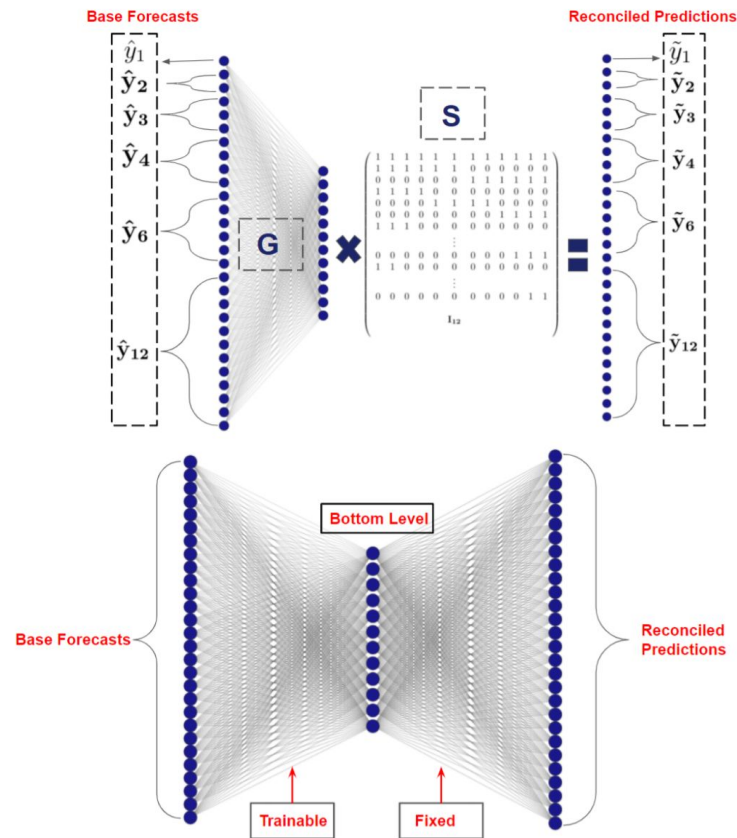


G is viewed as a trainable neural network in the form of an encoder

Methodology: NNs for Reconciliation part 2

Three different optimization methods:

1. S-UP12: Combine into the bottom level and Sum-UP
 - a. Similar with THieF
 - b. Model is optimized to produce bottom level predictions
 - c. **Coherence Ensured**
2. COH-28: Optimize across the whole hierarchy
 - a. Frozen - Non Trainable Decoder
 - b. Fixed weights equal to Summing Matrix S
 - c. Global Loss optimization
 - d. Hard constraint regularizes training
 - e. **Coherence Ensured**
3. Encoder - Decoder (ED): Trainable Decoder
 - a. Trainable Decoder (unfrozen weights)
 - b. **Coherence not Ensured**



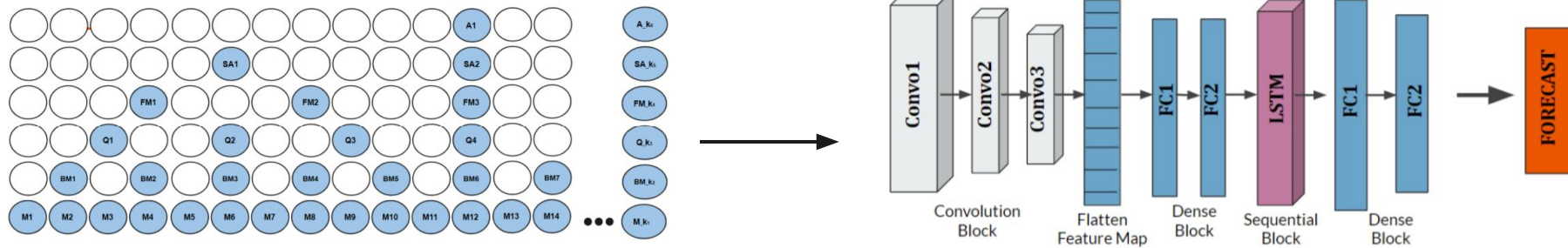
Methodology: Base Forecasts

Benchmark: **Exponential Smoothing (ETS)**

- Fitted on a single hierarchical level
- Easily optimized
- Good overall performance (Aggregation levels of higher frequency)

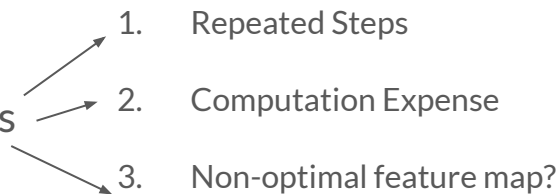
CNN - LSTM Architecture: Consider information from each hierarchical level

- 2D CNNs: Model the local relationships between aggregation levels
- LSTMs: Long - Term Patterns
- Enhancing the overall accuracy of base forecasts??



Methodology: Joint Predictions

Limitations of the defined Architecture:

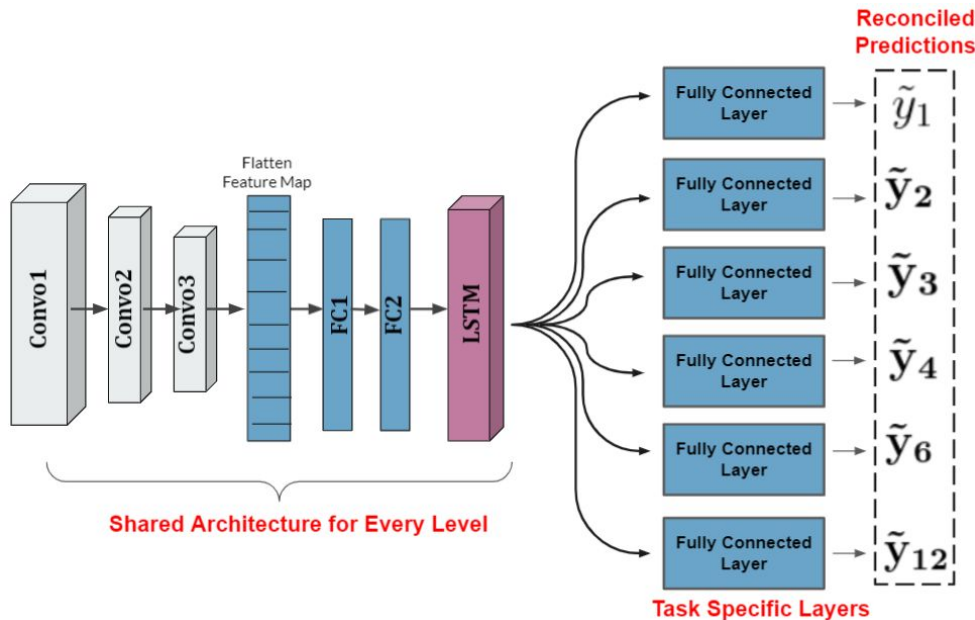
- Time consuming process
 - CNNs from individual models (probably) extract identical features
- 
1. Repeated Steps
 2. Computation Expense
 3. Non-optimal feature map?

Benefits of different defined approaches

- Modelling of Local Relationships (CNN-LSTM Architecture)
- Optimize across the hierarchy (COH-28)
- Training Regularization (COH-28)
- Map bottom level into full hierarchy (S-UP12)

Combine models for Base Forecasts & Reconciler into a joint framework trained end-to-end

Deep Temporal Hierarchies (DTH)



Features from CNN - LSTM Architecture

- Local Relationships
- Long-term patterns

Multi - Task model for predictions:

Optimizing each level regularizes the optimization of the others (CNN extracts Important Features)

Two Different Usages:

1. Use all reconciled Predictions (Small reconciliation loss)
2. Keep bottom predictions & S-UP (Ensures Coherence)

Experimental design

1. **Data** : M4 Dataset → 2 samples of Monthly Time Series

2. Fitting the Reconcilers :

a. **Per Series**: A unique reconciler on a single time series

- i. Sliding window
- ii. Evaluation on 88 time series

b. **Per Dataset**: Global Training on Multiple Series

- i. 15000 Time series
- ii. 15-fold Cross validation
- iii. Ensembles of 15 Reconcilers

3. Evaluation : **Mean Absolute Scaled Error (MASE)**

- a. Bottom Level
- b. Full hierarchy (28 stacked predictions)

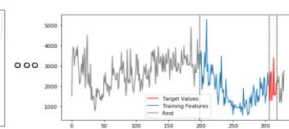
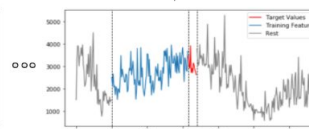
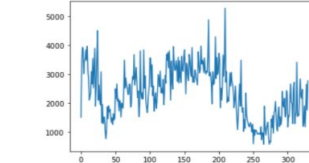
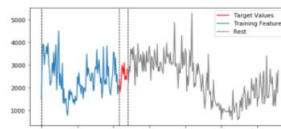
4. Comparison Experiments

a. **Individual Components**

- i. Compare the Reconcilers (Base Forecasts: **ETS**)
- ii. Compare Base Forecasts (ETS and CNN-LSTM)

b. **Joint performance of DTH**

Step1:
Rolling - Origins for building the data
points of the training set
Last split is kept for test set



Benchmarks A

1. ETS (non-reconciled predictions)
2. THieF (with structural scaling)

Benchmarks B

3. CNN - LSTM (individual forecasts)
4. CNN - LSTM (with COH-28)
5. CNN - LSTM (with S-UP12)
6. CNN - LSTM (with Structural Scaling)
7. ETS (with COH-28)
8. ETS (with S-UP12)



Mean is influenced by Outliers

It is included to highlight which model fail on particular series

Results: Individual Components

Reconcilers: Competitive with THieF

- COH-28 and ED **adequate performance** when fitted on a single series
- S-UP12 **fails** on fewer data (per-series)
- Increased performance when fitted globally (**outperform** THieF , ETS)
- COH-28 **best** performance overall

Reconcilers Comparison

Model	Per Series				Per Dataset			
	Bottom Level		Full Hierarchy		Bottom Level		Full Hierarchy	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
ETS	53.32	1.378	2.201	0.742	7.34	1.365	9918.6	0.232
THieF	29.98	1.287	2.14	0.626	79.34	1.355	28.195	0.201
S-UP12	16.75	2.8	1.65	1.01	7.231	1.336	8.513	0.186
COH-28	11.318	1.40	1.125	0.628	6.817	1.299	19.025	0.182
Encoder-Decoder (ED)	8.88	1.41	1.16	0.65	5.718	1.313	43.985	0.183

Why?

- Coherence??
- Optimization across the full hierarchy??
- Both??

Base Forecasts:

- CNN-LSTM **outperforms** ETS on every level
- The architecture could be used to **forecast series of lower frequencies** (fewer data points)

Base Forecast Comparison

Hierarchical Level	ETS		CNN-LSTM	
	Mean	Median	Mean	Median
Monthly	52.12	2.26	3.7	2.10
Bi-Monthly	1165.97	21.99	2.63	1.76
Quarterly	26.57	14.92	3.03	1.913
Four-Months	17.12	8.73	2.29	1.04
Semi-Annual	54.94	9.27	13.78	1.81

Results: Joint Predictions

Deep Temporal Hierarchies produces the best results

- Local Relationships ✓
- Long Term Dependencies between features ✓
- Regularized training ✓

- CNN-LSTM with COH-28 for Reconciliation performs equally well, but...
- Base Forecasts affect the overall performance :
Reconciling accurate forecasts.
- COH-28 outperforms Structural Scaling and S-UP12 as a reconciliation method

Family	Model	Bottom Level		Full Hierarchy	
		Mean	Median	Mean	Median
Individual Models	CNN-LSTM	3.7	2.10	0.25	0.22
	ETS	52.12	2.26	147.65	1.14
ETS w. reconciliation	ETS (w. COH-28)	12.146	7.157	0.83	0.693
	ETS (w. S-UP12)	13.195	8.028	0.889	0.732
	ETS (w. Struc)	876.15	13.07	147.60	1.10
CNN-LSTM w. reconciliation	CNN-LSTM (w. COH-28)	2.33	1.51	0.164	0.128
	CNN-LSTM (w. S-UP12)	4.409	1.998	0.325	0.207
	CNN-LSTM (w. Struc)	4.64	2.258	0.342	0.21
Joint Framework	DTH	2.12	1.46	0.133	0.171

Findings:

- Using Deep Learning to design the individual components of THieF produces competitive performance
- Combining them into a joint framework trained end-to-end increases the overall accuracy

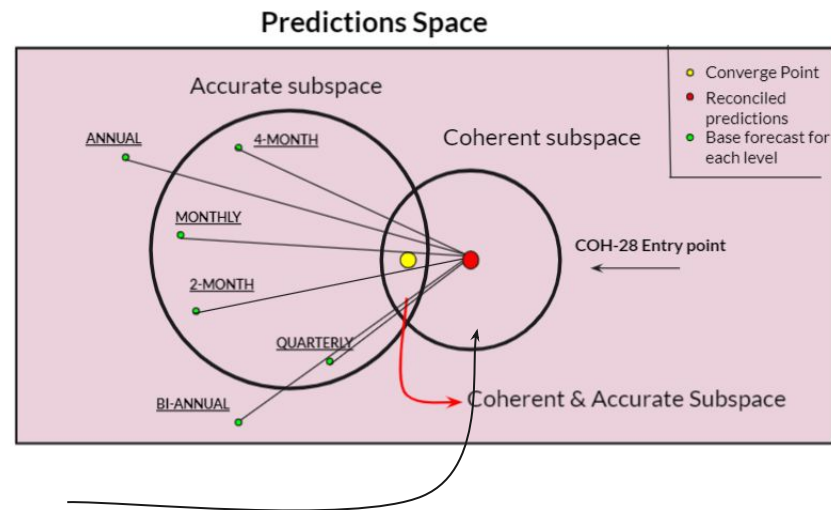
COH-28 vs S-UP12. Stepwise and Regularizer

Why COH-28 is more accurate?

1. Coherence as a Stepwise (S-UP12)
2. Coherence as a Regularizer (COH-28)
3. No Coherence (Encoder - Decoder)

Coherence as a hard constraint (frozen weights) is beneficial in terms of accuracy!!

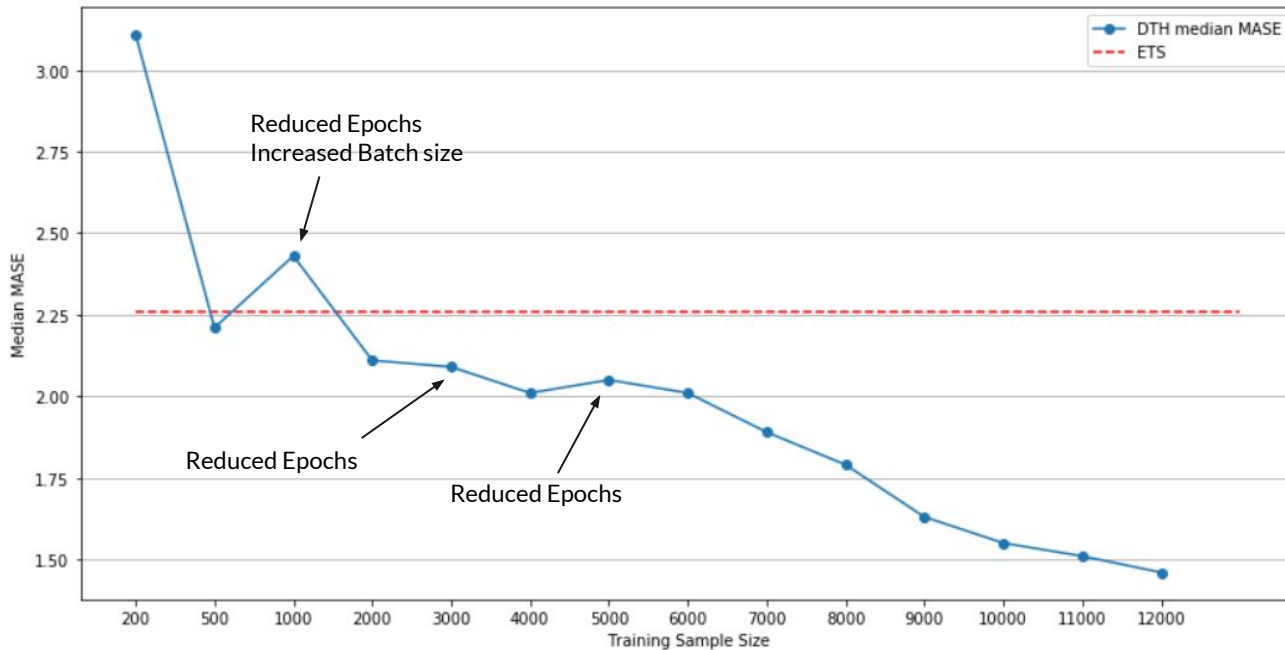
- Regularizes predictions on each level
- Less weights to be estimated
- Limits the total number of weight combinations available.
- Alternative: Keep bottom level and S-UP



Using coherence constraints to optimize training **uncovered new subspaces in the prediction space**

How much is enough?

Performance of DTH Relative to its Training Sample Size



DTH accuracy is relative good even when trained on fewer time series

Conclusions

Using Deep Learning to Forecast with Temporal Hierarchies - **Global Training** -

Neural Networks can be used **as an alternative** to well-established statistical methods

- For Base Forecasting
- For Reconciliation
- End-to-End

Coherence **as a regularizer** is beneficial in terms of accuracy

Future work:

1. Investigate minimum number of observations per series
2. Architecture Alternatives (Reconciliation & Full Framework)
3. Coherence as a regularizer



Thank you for your Attention!!
:)

Questions?

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